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The Future of the SETI Post-Detection Protocols: Progress Towards Revisions

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Brief History of Protocol Development

- 1989 IAA SETI Committee adopted 'Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence' (the Declaration).
 - By 1992 approved by IAA, IISL, IAF, COSPAR, IAU
- 1995 Draft 'Declaration of Principles Concerning Sending Communications with Extraterrestrial Intelligence' (the Reply Principles)
 - 1996 IAA Board of Trustees formal IAA Position Paper (revised 2007)
 - June 2000 presented to UN COPUOS
 - Report of COPUOS was approved by UNGA
- 2010 Revised 'Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence'
- IAA SETI Committee 2017, 2018, 2022
 - Task Force – current authors + Lori Walton, Andrew Siemion

Partial List of Current SETI Activities

- Resurgence with Breakthrough Listen and SETI Institute donations
- New state-of-the-art radio telescopes
 - Five-hundred-metre Aperture Spherical Telescope (FAST) in China
 - SKA precursors and pathfinders including MeerKAT array of sixty-four 13.5-metre diameter dishes (RSA), LOFAR (NL/EU), Murchison Widefield Array (AUS), and uGMRT (IND)
 - Canadian Hydrogen Intensity Mapping Experiment (CHIME) located in British Columbia
- Groups working on human factors
 - SETI Post-Detection Hub at the University of St. Andrews in Scotland launched by IAA SETI Committee members John Elliott and Martin Dominik
 - International Institute of Space Law Working Group on SETI and Law, initial convenor Michael Bohlander, now under the leadership of Andrea Harrington

Increase in conferences, symposia and other meetings

- biannual Astrobiology Science Conference
- NASA Astrobiology Program workshop “Communicating Discoveries in the Search for Life in the Universe”
- “Breaking News: We Found Extraterrestrial Life!” workshop
- Kavli Foundation/IAU Symposium (IAUS 387) “(Toward) Discovery of Life Beyond Earth and its Impact”
- Penn State SETI Symposia
- European Astrobiology Network Association (EANA) 2024 Conference
- Breakthrough Foundation Breakthrough Discuss conferences
- SETI Institute ongoing activities

Educational institution programmes in astrobiology include

- Australian Centre for Astrobiology, University of New South Wales
- University of Manchester
- Arizona Astrobiology Center (AABC), University of Arizona
- University of Edinburgh
- University of Tokyo
- Pierre and Marie Curie University
- Centro de Astrobiología (CAB) in Spain
- Arizona State University
- University of Washington
- University of California at Berkeley
- University of Chicago
- Nordic Network of Astrobiology Graduate Schools in Scandinavia

Draft Revised Post-Detection Declaration of Principles

- Hereby submitted for discussion and comment by the IAA SETI Committee and the SETI community
- Intention to have a finalised document to present to the IAA SETI Committee for consideration and approval at the 76th International Astronautical Congress in Sydney in October 2025
- Will be submitted to the IAA Board of Trustees for endorsement
- The method and process for the discussion and comments are to be determined by the IAA SETI Committee executives

Draft Revised Declaration Highlights

- Updates the current 2010 text
- Provides clarity in identifying the parties to the document
- Discusses engagement with social media
- Includes an express reference to technosignatures
- Commits the IAA SETI Committee to re-establishing a qualified Post-Detection Sub-Committee, while reserving the details for future discussion
- Does not address all the issues identified and described in our previous 2023 IAC paper in Baku
- Defers the issues regarding a global reply to the detection of confirmed extraterrestrial communication
- Does not address the separate and distinct subject of messaging to extraterrestrial intelligence in advance of a confirmed detected extraterrestrial signal (METI)

- The language in the Draft Revised Declaration is a compromise in certain places, in order to facilitate the release of the document for consideration and robust discussion, and should not be interpreted as a blanket endorsement by any of the authors.
- With this presentation of the Draft Revised Declaration, the work of the Task Group is completed. We thank all those who participated in the efforts to update the Protocols.

Article by Article Comparison of 2010 Declaration and 2024 Draft Revisions

- **2010 Preamble:** The parties to this declaration are individuals and institutions participating in the scientific Search for Extraterrestrial Intelligence (SETI).
- The purpose of this document is to declare our commitment to conduct this search in a scientifically valid and transparent manner and to establish uniform procedures for the announcement of a confirmed SETI detection.
- This commitment is made in recognition of the profound scientific, social, ethical, legal, philosophical and other implications of a SETI detection. As this enterprise enjoys wide public interest, but engenders uncertainty about how information collected during the search will be handled, the signatories have voluntarily constructed this declaration. It, together with a current list of signatory parties, will be placed on file with the International Academy of Astronautics (IAA).

2024 Proposed Revised Preamble

- Expressly defines scope: search for ‘technosignatures’ or evidence of past or present intelligent life and technology beyond Earth
- Defines purposes, including:
 - balancing the community imperative of providing rapid and accurate and a Code of Conduct that will be periodically re-examined and updated information to a wide audience
 - with consideration for the safety and exposure of individual scientists
 - and to proactively inform the public of the procedures and guidelines
- Recognises that the scientific and communications landscapes are ever-changing, and the Declaration will be supplemented with Best Practices Guidelines

2010 Text Articles 1 and 2:

1. Searching:

- SETI experiments will be conducted transparently, and its practitioners will be free to present reports on activities and results in public and professional fora. They will also be responsive to news organizations and other public communications media about their work.

2. Handling candidate evidence:

- In the event of a suspected detection of extraterrestrial intelligence, the discoverer will make all efforts to verify the detection, using the resources available to the discoverer and with the collaboration of other investigators, whether or not signatories to this Declaration. Such efforts will include, but not be limited to, observations at more than one facility and/or by more than one organization. There is no obligation to disclose verification efforts while they are underway, and there should be no premature disclosures pending verification. Inquiries from the media and news organizations should be responded to promptly and honestly. Information about candidate signals or other detections should be treated in the same way that any scientist would treat provisional laboratory results. The Rio Scale, or its equivalent, should be used as a guide to the import and significance of candidate discoveries for the benefit of non-specialist audiences.

2024 Revised draft puts handling of candidate evidence before searching

2024 Proposed Text:

1. Handling Candidate Evidence

- In the event of a putative detection of extraterrestrial intelligence, the discoverer will make all efforts to verify the detection, using the resources available to the discoverer and in collaboration with other investigators. Such efforts will include, but not be limited to, observations at more than one facility and/or by more than one organisation.
- Information about candidate signals or other potential detections should be handled with the same care as provisional laboratory results, **recognising that** initial findings may be incomplete or ambiguous, requiring thorough analysis and confirmation, which could be a lengthy process. **It is** crucial to uphold the highest standards of scientific responsibility and integrity **throughout this process**. **Additionally**, best practices in science communication should be employed **to clearly convey the importance and significance of candidate discoveries to non-specialist audiences**.
- Reference to Rio Scale deleted

2. Searching:

- SETI practitioners and their institutions and organisations will be free to present reports on activities and results in public and professional fora. Institutions and organisations should be responsive to news organisations, social media platforms, and other public communications media. Responses to inquiries should be prompt, accurate and honest.
- There is no obligation to disclose verification efforts until a discovery is confirmed. If a candidate technosignature is discovered, communication about ongoing observations and analyses may be necessary to dispel rumours and provide accurate and reliable information.
- Speculative or unconfirmed conclusions should be clearly identified as such, and exaggerated statements should be avoided.

2. Searching (continued):

- Institutions and organisations should engage with news organisations to provide accurate and timely updates of their activities, ensuring proactive engagement with the public to provide a source for accurate information and consider other ways to explain a putative signal among public audiences.
- Individual practitioners should have the right to decline from engaging directly or continually with the media, including social media if they choose to do so, and shall use their best efforts to ensure that their organisation or institution provides updates on their science.

2010 Text Article 3. Confirmed detections:

- If the verification process confirms – by the consensus of the other investigators involved and to a degree of certainty judged by the discoverers to be credible – that a signal or other evidence is due to extraterrestrial intelligence, the discoverer shall report this conclusion in a full and complete open manner to the public, the scientific community, and the Secretary General of the United Nations. The confirmation report will include the basic data, the process and results of the verification efforts, any conclusions and interpretations, and any detected information content of the signal itself. A formal report will also be made to the International Astronomical Union (IAU).

2024 Proposed Text Article 3. Confirmed detections:

- If the verification process confirms – by the consensus of the other investigators involved and to a degree of certainty judged by the discoverers to be credible – that a signal or other evidence is due to extraterrestrial intelligence, the discoverer shall report this conclusion in a full, complete and open manner to the public, the scientific community, and the Secretary General of the United Nations.
- The confirmation report should include the basic data, the data analysis process and results of the verification efforts, any conclusions and interpretations, and any detected information content of the signal itself. A formal report will also be made to the International Academy of Astronautics, the International Astronomical Union Central Bureau for Astronomical Telegrams, the United Nations Secretary General, the Committee on Space Research (COSPAR) of the International Science Council, the International Institute of Space Law, and the International Telecommunications Union.

2010 Text Article 4:

- All data necessary for the confirmation of the detection should be made available to the international scientific community through publications, meetings, conferences, and other appropriate means.

2024 Proposed Text Article 4: Data Accessibility:

- All data necessary for the confirmation of the detection to be replicated should be made available to the international scientific community through publications, meetings, conferences, official websites, and other appropriate means.

2010 Text Article 5:

- The discovery should be monitored. Any data bearing on the evidence of extraterrestrial intelligence should be recorded and stored permanently to the greatest extent feasible and practicable, in a form that will make it available to observers and to the scientific community for further analysis and interpretation.

2024 Proposed Text Article 5: Monitoring and Archiving:

- The discovery should be monitored and best practices for the safe, reliable and resilient handling of data should be employed. Any and all data bearing on the evidence of extraterrestrial intelligence should be recorded and stored permanently to the greatest extent feasible and practicable, in a form that will make it available to observers and to the scientific community for replication of results and further analysis.

2010 Text Article 6:

- If the evidence of detection is in the form of electromagnetic signals, observers should seek international agreement to protect the appropriate frequencies by exercising the extraordinary procedures established within the World Administrative Radio Council of the International Telecommunication Union.

2024 Proposed Text Article 6: Data and Frequency Protection:

- If the evidence of detection is in the form of electromagnetic signals, observers should seek international agreement to protect the appropriate frequencies by exercising the extraordinary procedures established within the International Telecommunication Union. Evidence in a form other than electromagnetic signals should be protected utilising best scientific practices.

2010 Text Article 7: Post Detection:

- A Post-Detection Task Group under the auspices of the IAA SETI Permanent Study Group has been established to assist in matters that may arise in the event of a confirmed signal, and to support the scientific and public analysis by offering guidance, interpretation, and discussion of the wider implications of the detection.

2024 Proposed Text Article 7: Post-Detection Protocol:

- The IAA SETI Committee will appoint a Post-Detection Sub-Committee to assist in matters that may arise in the event of a confirmed signal, and to support the scientific and public analysis by offering guidance, interpretation, and discussion of the wider implications of the detection. This sub-committee will be chaired by the chair of the IAA SETI Committee and should include communications professionals.

2024 Proposed Text Article 7 (continued):

- The SETI Committee Executive Officers will ***support and assist*** the Post-Detection Sub-Committee ***with engagements with social media platforms and news organisations*** to responsively and effectively assist in the dissemination of accurate and reliable information.
- The IAA SETI Committee ***will maintain and support working groups dedicated to Post-Detection issues***, providing assistance during the discovery process, aiding in scientific analysis, and ***focusing on subjects such as best practices for public communication*** about technosignatures science. These working groups will ***actively collaborate with interdisciplinary researchers*** to explore the broader implications of potential discoveries.

2010 Text Article 8: Response to signals:

- In the case of the confirmed detection of a signal, signatories to this declaration will not respond without first seeking guidance and consent of a broadly representative international body, such as the United Nations.

2024 Proposed Text Article 8: Communications with ETI following a confirmed detection:

- **Signatories to this Declaration** will refrain from making any unilateral communications in response to a confirmed extraterrestrial signal, **and** shall cooperate with appropriate international consultations. **These consultations should be conducted through a broadly representative international body, such as the United Nations.** The specific procedures for such consultations will be outlined in a separate agreement, declaration, or arrangement to ensure a coordinated and responsible approach.
 - NOTE: This Declaration is not addressed to the separate and distinct subject of METI

2024 Proposed Text Article 9: Ethical and Legal Considerations:

- SETI practitioners shall adhere to the highest ethical standards, ensuring cooperation, honesty, and integrity in all aspects of their work. They will collaborate with international legal authorities to establish clear frameworks for managing the dissemination of information about potential extraterrestrial detections, ensuring compliance with relevant laws and regulations. SETI practitioners shall cooperate in establishing ethical standards for the handling of any putative and confirmed detected evidence of extraterrestrial intelligence.

Thank You

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